

Problem Description:

Mr. Alex just installed new bands in his feet and they do not feel like they should, making it where he cannot stop!

However the Alex cannot do the math while stopping himself from crashing into things. Using  $M = -dx$ , please help him find out the M, d, or x value as he requests it.

Constraints:

M, D, X = 0 to 10000 (or a question mark).

Input Format:

Each line will start capital letters only, then the value of that variable. The value must be a decimal number.

Output Format:

Print the output in a separate lines contains the variable, a space, then the value.

Logical Test Cases

Test Case 1

INPUT (STDIN)

M 14.00  
D -4.80  
X ?

EXPECTED OUTPUT

x 2.92

Test Case 2

INPUT (STDIN)

M 12.00  
D -5.80  
X ?

EXPECTED OUTPUT

x 2.07

✓ Mandatory Test Cases

Test Case 1

KEYWORD

```
char var[3][LEN];
```

Test Case 2

KEYWORD

```
char inp[3][LEN];
```

✓ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

4

Test Case 2

TOKEN COUNT

320

Test Case 3

NLOC

30

Searching1.c +

```
1 #include <stdio.h>
2 #include <string.h>
3 #include <stdlib.h>
4
5 #define LEN 100
6
7 char var[3][LEN];
8 char inp[3][LEN];
9
10 int main() {
11     double m = 0, d = 0, x = 0;
12     int m_idx = -1, d_idx = -1, x_idx = -1;
13     for (int i = 0; i < 3; i++) {
14         if (scanf("%s %s", var[i], inp[i]) != 2) return 0;
15         if (strcmp(var[i], "M") == 0 || strcmp(var[i], "m") == 0) {
16             if (strcmp(inp[i], "?") != 0) m = atof(inp[i]);
17             else m_idx = i;
18         } else if (strcmp(var[i], "D") == 0 || strcmp(var[i], "d") == 0) {
19             if (strcmp(inp[i], "?") != 0) d = atof(inp[i]);
20
21         } else if (strcmp(var[i], "X") == 0 || strcmp(var[i], "x") == 0) {
22             if (strcmp(inp[i], "?") != 0) x = atof(inp[i]);
23             else x_idx = i;
24         }
25     }
```

main | CH.SC.U4CSE25225 | Searching1.c

Ln 38, Col 1 Spaces: 4 UTF-8 CRLF C

```
26  if (x_idx != -1) {
27      x = m / (-d);
28      printf("x %.2f\n", x);
29  } else if (d_idx != -1) {
30      d = -m / x;
31      printf("d %.2f\n", d);
32  } else if (m_idx != -1) {
```

```
33     m = -d * x;  
34     printf("m %.2f\n", m);  
35 }  
36 return 0;  
37}  
38
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
M 14.00
```

```
D -4.80
```

```
X ?
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching1.exe
```

```
x 2.92
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
M 12.00
```

```
D -5.80
```

```
X ?
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching1.exe
```

```
x 2.07
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```